

ABSTRACT

The smart camera project presented in this research aims to develop an intelligent surveillance system capable of detecting and analysing human activities in real-time. Leveraging state-of-the-art image processing techniques and machine learning algorithms, the system focuses on enhancing security, surveillance, and safety in diverse environments. The smart camera is equipped with high-resolution sensors and embedded hardware capable of processing and analysing video streams efficiently. It utilises computer vision algorithms, including object detection, tracking, and activity recognition, to identify and categorise various human activities. The system's robustness is achieved through extensive training using labeled datasets to enable accurate and reliable activity recognition. Deep learning models are employed to generalise and adapt to different environmental conditions, ensuring versatility and adaptability.

TABLE OF CONTENTS

ACKNOWLEDGEMENT	i
ABSTRACT	ii
TABLE OF CONTENTS	iii
LIST OF FIGURES	v
LIST OF TABLES	vi
1. Introduction	1
2. Literature Survey	2
3. System Analysis	5
3.1 Existing system	5
3.1.1 Disadvantages of Existing System	6
3.2 Proposed System	7
3.2.1 Advantages of Proposed System	8
4. System Study	9
4.1 Feasibility Study	9
4.1.1 Economical Feasibility	10
4.1.2 Technical Feasibility	10
4.1.3 Social Feasibility	11
5. Hardware and Software requirements	12
5.1 Hardware requirements	13
5.2 Software requirements	14
6. Architecture	16
7. Diagrams	19
7.1 Dataflow diagram	
8. Screen Shots	20
9. Implementation	22
10. Code	23
11. Testing	29
12. Conclusion	36
13. Reference	37
14. Gann Chart	38

LIST OF FIGURES

Figure No.	Particulars	Page no.
6.1	Flow chart	16
7.1	Dataflow diagram	19
8.1	Detect the person	20
8.2	No person detected	20
8.3	Full body detection	21
8.4	In case of animals	21

LIST OF TABLES

Table No.	Particulars	Page no.
2.1	Literature survey	4
11.1	Hardware Testing	32
11.2	Software Testing	32
11.3	Functionality Testing	33
11.4	Integration Testing	33
11.5	Performance Testing	34
11.6	Security Testing	34
11.7	Usability Testing	35

1. INTRODUCTION

In an era of advancing technology, smart cameras have emerged as versatile tools with the ability to transform various aspects of our lives. A specialised variant of these cameras, equipped with sophisticated object recognition algorithms, has been designed to focus solely on detecting human motion. These smart cameras, harnessing the power of artificial intelligence and computer vision, have the unique capability of discerning human movement from other activities in their surroundings.

The concept of a smart camera that exclusively detects human motion brings a myriad of applications and benefits to diverse industries and settings. By honing in on human activity, these cameras offer enhanced security, improved surveillance, and efficient monitoring systems while minimising false alarms triggered by non-human events.

With a sharp focus on privacy and ethical considerations, these cameras are tailored to respect individual rights while providing an added layer of protection. By only capturing human movements and omitting other forms of activity, they offer a non-intrusive solution that fosters a sense of safety and privacy for users.

2. LITERATURE SURVEY

Surveillance Systems using CCTV cameras or any other surveillance devices record the footage all the time they are active. Most of the Data recorded is idle data where no activity takes place. When an activity which has occurred is supposed to be viewed the user has to go through all of the footage to check when and what had happened. This system eases this time consuming task. It uses Action Recognition to filter the idle movement data and trims the part where activity has been recorded using movement detection and various actions. It has a prominent scope for utilization in making the process of examining the footage from surveillance systems simple. This project could be implemented at public places and could enable respective concerned departments to investigate the data at an amazingly rapid rate. Thus, in an era of increasing crimes rates, this project could be a breakthrough in untangling the evidence data and eventually assist in increasing the rate of crime resolution

Approaches	Categories	Purpose	Features	Algorithms	Characteristics	
					Advantages	Limitations
MI SMART CAMERA	Smart camera	These cameras are designed primarily for home security surveillance.	They usually offer features like motion detection, night vision, two-way audio communication, and smartphone integration for remote monitoring..	functions like motion detection and video processing are proprietary to Xiaomi and may involve techniques such as computer vision and machine learning for enhanced performance.	Affordable pricing Integration with Xiaomi ecosystem Ease of availability	Limited advanced features Potential privacy concerns
Deep Sentinal	Smart camera.	provide proactive home security by combining advanced technology with human intervention to deter potential threats and respond quickly to suspicious activity.	Live video monitoring AI-powered threat detection Human intervention Two-way audio	designed to analyze camera feeds in real-time, detecting and identifying potential threats based on factors such as motion patterns, object recognition, and behavioral analysis.	Proactive security Reduced false alarms Peace of mind	Cost Dependence on internet and power Coverage limitations

Nest Cam IQ	Smart camera	The Nest IQ cameras are designed to provide advanced home security monitoring with features like high-resolution video, intelligent alerts, and integration with other smart home devices.	Facial recognition Intelligent alerts Two-way audio Integration	advanced algorithms for motion and sound detection, facial recognition, and image processing to provide intelligent alerts and enhance overall performance.	High-resolution video Facial recognition Integration with other smart home devices allows for a comprehensive home security system.	Cost Subscription Dependence on internet connection
Arlo Smart	Smart camera	arlo camera models, including indoor and outdoor cameras, as well as video doorbells. The service is available in different subscription tiers, offering varying levels of features and benefits.	Intelligent alerts Cloud storage Activity zones Person detection Rich notifications	motion detection, person detection, and activity zone customization are proprietary to Arlo Technologies	Enhanced security Convenience Customization	Subscription cost Dependence on internet connection Compatibility
Ring	Smart camera	It aims to empower users to monitor and protect their homes remotely, deter potential intruders, and enhance overall safety and peace of mind.	Live Video Streaming Motion Detection Cloud Storage Integration	algorithms are designed to analyze video footage and trigger alerts based on predefined criteria.	Affordability Easy Installation Scalability Community Sharing	Dependence on Internet Connection Privacy Concerns Subscription Costs

Table 2.1:*Literature Survey*

3. SYSTEM ANALYSIS

System analysis is a critical phase in the project management and development process. It provides a solid foundation for project planning, design, development, and implementation, helping to ensure that the final system meets the intended objectives and user needs. This analysis aims to gather detailed information about the existing system and its disadvantages, proposed system and its attributes like its advantages and advancements.

3.1 EXISTING SYSTEM

The CCTV records everything and due to that lot of storage is consumed. When the crime happens we have to go through all the footage which takes a lot of time. CCTV (CLOSED – CIRCUIT – TELEVISION) is a monitoring system comprised of recording devices that are not publicly distributed. Surveillance cameras are directly connected to private monitors and recording devices for immediate viewing and storage. The primary use is for home or office security. Disadvantages of CCTV is the cost. Cameras, monitors, recording devices, and other equipment costs start to add up. One has to monitor in screen what is happening around the surveillance. The existing smart camera represents the culmination of years of technological innovation, boasting an array of advanced features and capabilities. With high-resolution sensors, it captures detailed footage in various lighting conditions, ensuring clarity and precision. Its sophisticated AI algorithms enable intelligent motion detection, filtering out false alarms and providing accurate alerts. Integration with smart home systems allows seamless control and monitoring from anywhere via smartphone apps or voice commands. Cloud storage ensures secure and convenient access to recorded footage.

3.1.1 DISADVANTAGES OF EXISTING SYSTEM

1. **Privacy Concerns:** The advanced capabilities of smart cameras raise privacy issues, as they can collect and process sensitive data, leading to potential misuse if not handled responsibly.
2. **Initial Cost:** Smart cameras often come with higher upfront costs compared to traditional surveillance systems, making them less accessible for some users.
3. **Dependence on Connectivity:** Smart cameras may rely on an internet connection or cloud services for some features, which could be problematic in areas with limited or unstable internet access.
4. **False Alarms:** The AI-based analytics in smart cameras can sometimes lead to false alarms, triggering responses for non-threatening events, which can be frustrating for users and lead to inefficiencies.
5. **Vulnerability to Hacking:** Like any connected device, smart cameras are susceptible to hacking and unauthorised access, making security measures and regular updates crucial to mitigate risks.

3.2 PROPOSED SYSTEM

For a smart camera project to detect human activities, we use computer vision algorithms to analyse video feed and recognise actions like walking, running, or gestures. Incorporate deep learning models like CNNs or RNNs for activity recognition. Implement motion detection, object tracking, and event triggers to capture relevant activities. Ensure the system is scalable, efficient, and has proper privacy measures in place. Additionally, consider integrating cloud storage and real-time alerts for enhanced functionality. Our software helps in solving those problems. It only records when there is any human activity. It stops recording after 5sec. This helps in reducing storage and time.

The proposed smart camera represents a paradigm shift in surveillance technology, focusing on efficiency, autonomy, and enhanced functionality. Unlike its predecessors, this innovative camera is designed to significantly decrease storage requirements while still delivering reliable performance. By leveraging advanced compression algorithms and onboard processing power, it minimizes the size of recorded footage without compromising on quality. Moreover, it operates seamlessly without the need for an internet connection, ensuring continuous monitoring even in remote locations or during network outages.

One of the standout features of this proposed smart camera is its sophisticated body movement detection capabilities. This advanced detection technology not only reduces false alarms triggered by non-human activity but also enhances security by focusing specifically on human presence.

In essence, the proposed smart camera represents a leap forward in surveillance technology, offering unparalleled efficiency, autonomy, and precision in detecting human movement while simultaneously minimizing storage requirements and operating independently of internet connectivity.

3.2.1 ADVANTAGES OF PROPOSED SYSTEM

1. **Enhanced Security:** Smart cameras can detect and analyse suspicious activities, providing real-time alerts and ensuring better security for homes, businesses, and public spaces.
2. **Intelligent Analytics:** With built-in AI and computer vision capabilities, smart cameras can identify and track objects, recognise faces, and analyse data, offering valuable insights and reducing the need for manual monitoring.
3. **Cost-Effectiveness:** These cameras can reduce operational costs by automating tasks, optimising resources, and minimising the need for human intervention.
4. **Real-time Response:** Smart cameras can respond instantly to events, allowing for quick actions in critical situations and enhancing overall safety and efficiency.
5. **Integration with IoT:** Smart cameras can seamlessly integrate with other smart devices and Internet of Things (IoT) systems, creating a more interconnected and efficient ecosystem.

4. SYSTEM STUDY

System Study is the process of collecting and interpreting facts, identifying the problems, and decomposition of a system into its components. It is a problem solving technique that improves the system and ensures that all the components of the system work efficiently to accomplish their purpose .Design is a meaningful engineering representation of something that is to be built. It is the most crucial phase in the developments of a system. Software design is a process through which the requirements are translated into a representation of software.

Design is a place where design is fostered in software Engineering. Based on the user requirements and the detailed analysis of the existing system, the new system must be designed. This is the phase of system designing. Design is the perfect way to accurately translate a customers requirement in the finished software product. Design creates a representation or model, provides details about software data structure, architecture, interfaces and components that are necessary to implement a system. The logical system design arrived at as a result of systems analysis is converted into physical system design.

4.1 FEASIBILITY STUDY

The feasibility of the project is analyzed in this phase and a business proposal is put forth with a very general plan for the project and some cost estimates. During system analysis the feasibility study of the proposed system is to be carried out. This is to ensure that the proposed system is not a burden to the company. For feasibility analysis, some understanding of the major requirements for the system is essential.

Three key considerations involved in the feasibility analysis are:

- Economical Feasibility
- Technical Feasibility
- Social Feasibility

4.1.1 Economical Feasibility:

1. **Cost-Benefit Analysis:** Estimate the initial investment required for implementing the smart camera system, including hardware, software, installation, and training costs. Compare these costs against the expected benefits, such as increased security, reduced labor costs, or improved operational efficiency.
2. **Return on Investment (ROI):** Calculate the potential ROI by quantifying the expected cost savings or revenue gains resulting from the implementation of smart camera technology over a defined period.
3. **Total Cost of Ownership (TCO):** Consider not only the upfront costs but also ongoing expenses such as maintenance, upgrades, and support services to determine the total cost of ownership over the system's lifecycle.

4.1.2 Technical Feasibility

1. **Hardware and Software Requirements:** Evaluate whether the required hardware (cameras, processing units, storage) and software (image processing algorithms, AI models) are readily available and compatible with existing infrastructure.
2. **Integration with Existing Systems:** Determine if the smart camera technology can be seamlessly integrated with other systems or if significant modifications are needed.
3. **Scalability and Performance:** Assess whether the proposed system can scale to accommodate future growth in data volume or processing requirements while maintaining acceptable performance levels.
4. **Reliability and Security:** Consider whether the smart camera system can operate reliably under various conditions and if adequate measures are in place to protect data privacy and prevent unauthorized access.

4.1.3 Social Feasibility

- 1. Privacy Concerns:** Evaluate how the deployment of smart cameras may affect individuals' privacy rights. Consider factors such as the collection, storage, and use of personal data captured by the cameras, as well as the potential for unauthorized surveillance or data breaches.
- 2. Ethical Implications:** Consider the ethical implications of using smart cameras, such as the potential for discriminatory or biased outcomes in automated decision-making processes, as well as concerns about the erosion of civil liberties and the normalization of constant surveillance.
- 3. Trust and Transparency:** Establish trust with stakeholders by being transparent about the purpose, capabilities, and limitations of the smart camera system. Provide clear information about data handling practices, security measures, and mechanisms for addressing grievances or concerns.

5. HARDWARE AND SOFTWARE REQUIREMENTS

The hardware and software requirements outlined for the proposed system are pivotal elements that warrant careful consideration to guarantee the system's optimal functionality and high-performance capabilities. The following mentioned hardware and software requirements are those which is suggestable for the successful working and execution of the proposed system.

HARDWARE REQUIRED

- Laptop with camera / Desktop with webcam.
- 2 GB RAM minimum.
- Storage internal / external

SOFTWARE REQUIRED

- Python toolkit
- Visual studio IDE.
- OpenCV package

5.1 HARDWARE REQUIREMENTS

1. **Laptop/Desktop Computer:** A laptop or desktop computer with a built-in camera or an external webcam is necessary for capturing video footage. Ensure that the laptop/desktop meets the minimum system requirements for running the necessary software for smart camera functionalities.
2. **RAM (Random Access Memory):** A minimum of 2 GB of RAM is specified, but higher RAM capacity may be preferable for smoother performance, especially when processing high-definition video or running multiple applications simultaneously. For more demanding applications such as real-time object detection or facial recognition, consider higher RAM capacities to accommodate the computational requirements.
3. **Storage: Internal Storage:** The laptop/desktop should have sufficient internal storage capacity to store captured video footage and any processed data. The exact storage requirement will depend on factors such as the resolution and duration of the videos, as well as the amount of data generated by the smart camera application.

External Storage: Optionally, you may consider using external storage devices such as USB drives, external hard drives, or network-attached storage (NAS) systems to expand the storage capacity or to facilitate data backup and archiving.

5.2 SOFTWARE REQUIREMENTS

1. **Python Toolkit:** Python is a popular programming language for developing smart camera applications due to its simplicity, versatility, and extensive libraries for data processing and machine learning. Along with the core Python installation, you may want to install the Anaconda distribution, which includes commonly used libraries for scientific computing and data analysis.
2. **Integrated Development Environment (IDE):** Visual Studio Code (VS Code) is a lightweight yet powerful IDE that provides excellent support for Python development, including features like syntax highlighting, code completion, debugging, and version control integration. Alternatively, if you prefer a more feature-rich IDE, you can use PyCharm or Spyder, both of which offer robust Python development environments with advanced debugging and profiling tools.
3. **OpenCV (Open Source Computer Vision Library):** OpenCV is an open-source library for computer vision and image processing tasks. It provides a wide range of functionalities for manipulating images and videos, including algorithms for feature detection, object tracking, and camera calibration.

To install OpenCV for Python, you can use the pip package manager with the command: **pip install opencv-python**.

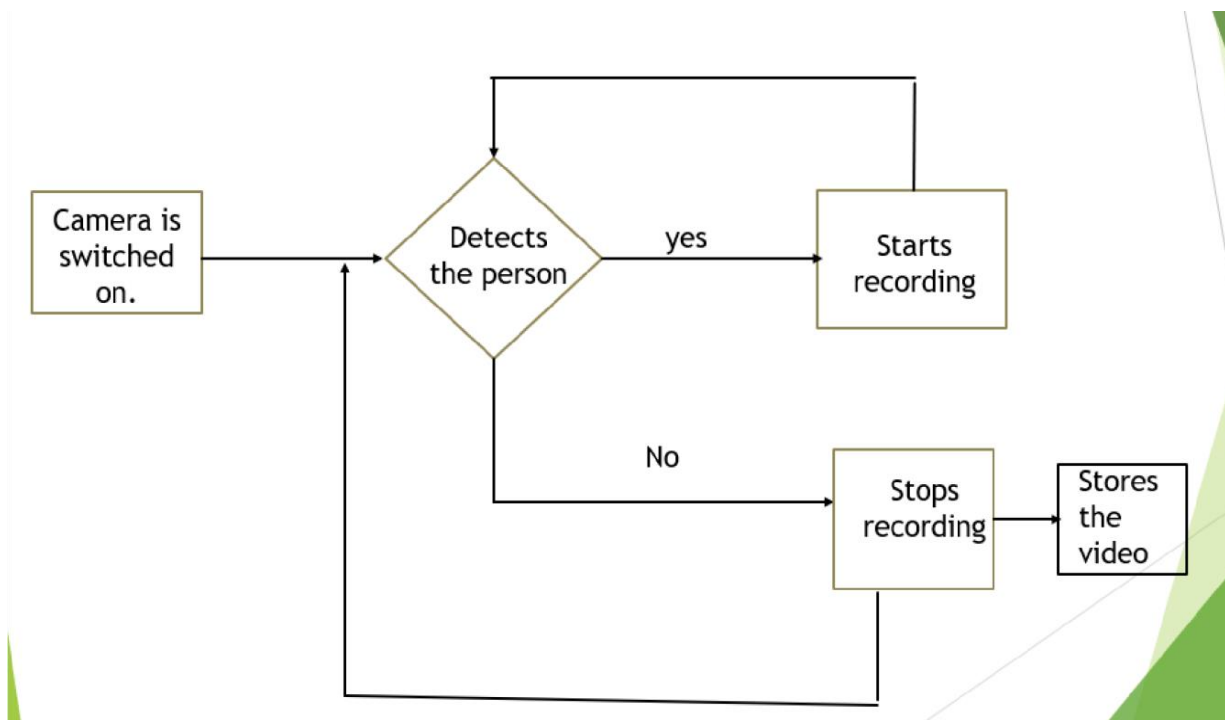
Additionally, consider installing the **opencv-contrib-python** package, which includes extra modules and features not available in the core OpenCV distribution. Python is a high-level, interpreted, interactive and object-oriented scripting language. Python is designed to be highly readable. It uses English words frequently whereas other languages use punctuation, and it has fewer syntactic constructions than other languages.

1. **Python is Interpreted:** Python is processed at runtime by the interpreter. You do not need to compile your program before executing it. This is similar to PERL and PHP.
2. **Python is Interactive:** You can actually sit at a Python prompt and interact with the interpreter directly to write your programs.
3. **Python is Object-Oriented:** Python supports Object-Oriented style or technique of programming that encapsulates code within object

6. ARCHITECTURE

Architecture typically refers to the structural and organizational framework or design of a system, software, building, or any complex project. The architecture provides a high-level overview of how the project is structured and how its various components or elements interact with each other. It outlines the fundamental design principles, components, and their relationships, and helps stakeholders understand the project's overall framework

Flowchart



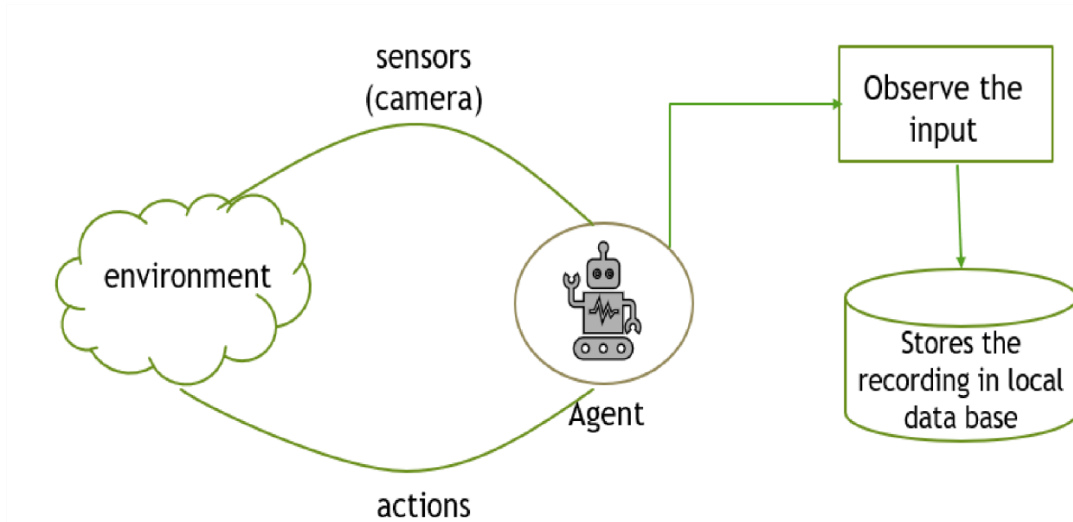
1. **Camera Activation:** The smart camera system is activated either manually or automatically. This could be triggered by a user command, a scheduled time, or a sensor detecting movement in the camera's vicinity.
2. **Detection of a Person:** Once the camera is activated, it continuously captures images or video frames using its sensors. The captured images or frames are processed using computer vision algorithms to detect the presence of a person within the camera's field of view. Detection algorithms analyze visual features such as shape, color, and motion to identify human subjects.
3. **Start Recording:** Upon detecting a person in the camera's view, the smart camera system initiates the recording process. Recording can be triggered automatically by the detection of a person, or it may require confirmation from the user or an automated decision-making process based on predefined criteria.
4. **Video Capture:** The smart camera system records video footage of the detected person, capturing a continuous stream of frames at a specified frame rate and resolution. Video recording continues until the person is no longer detected or until a predetermined duration or file size limit is reached.
5. **Storage of Recorded Data:** The recorded video data is stored in a designated storage location, either internally within the camera or externally on a connected storage device. Storage options may include onboard memory, removable memory cards, network-attached storage (NAS) devices, or cloud storage services. Recorded data may be stored in a compressed format to conserve storage space and facilitate efficient transmission and retrieval.

6.Metadata Generating :Along with the recorded video data ,the smart camera system generates metadata containing additional information about the recording, such as timestamps, and any relevant contextual data. Metadata helps organize and index recorded footage for easy retrieval and analysis

6. Stop Recording: Once the person is no longer detected within the camera's view or when the recording duration reaches its limit, the smart camera system stops recording. Recording can also be manually stopped by the user or through automated decision-making based on predefined criteria.

this sequence of actions, a smart camera system can effectively detect the presence of a person, initiate recording, store the recorded data, and provide users with relevant notifications or alerts as needed. This functionality is essential for applications such as surveillance, security monitoring, and activity tracking.

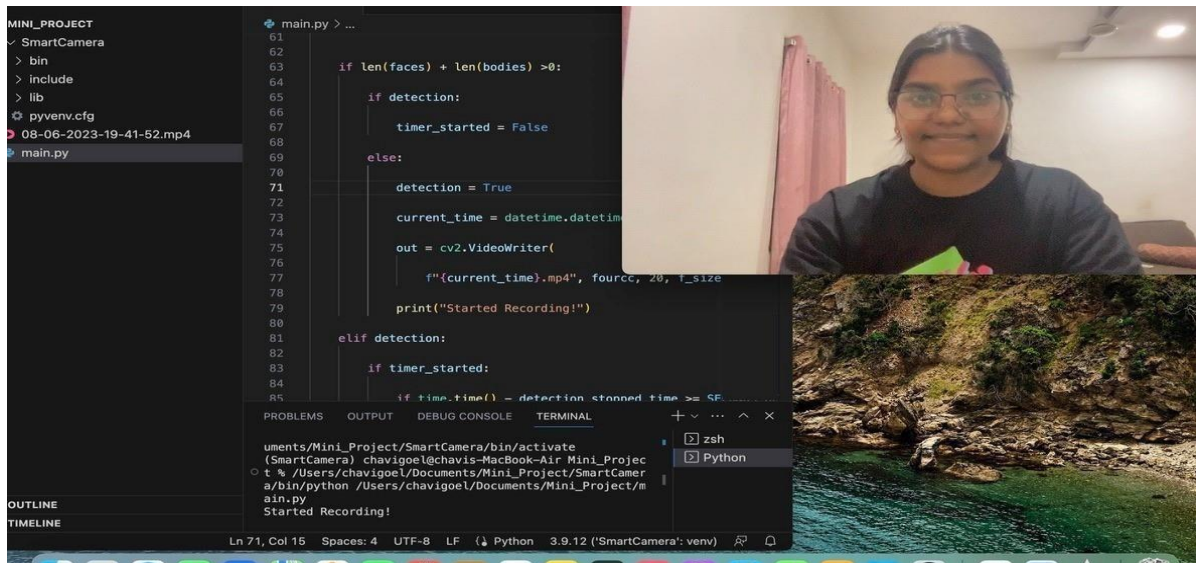
7. DIAGRAMS



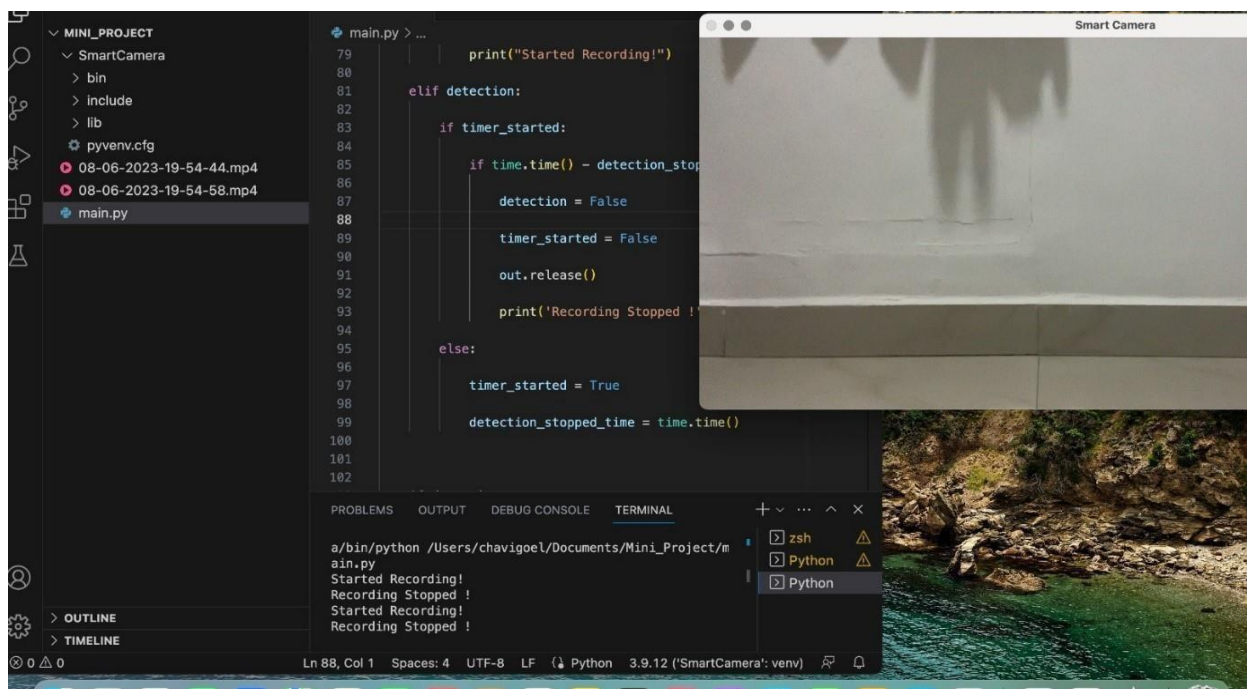
7..1 DATA FLOW DIAGRAM

In a storage-efficient smart camera system that records video selectively when a person is detected and operates seamlessly without data, the flow of information is orchestrated with efficiency. Initially, the camera captures continuous images or video frames. These data are then analyzed for person detection. Upon detecting a person in the frame, the recording process is triggered, and storage allocation is initiated. The camera promptly starts recording video, continuing until the person exits the frame or a predetermined duration elapses. During periods without person detection, the system remains operational without recording data, conserving resources effectively. Optionally, metadata is generated to provide additional context for the recorded footage. Subsequently, the recorded video, possibly compressed for optimal storage usage, is stored in the allocated space. This streamlined process ensures efficient storage management while maintaining seamless operation, enhancing the system's effectiveness and usability.

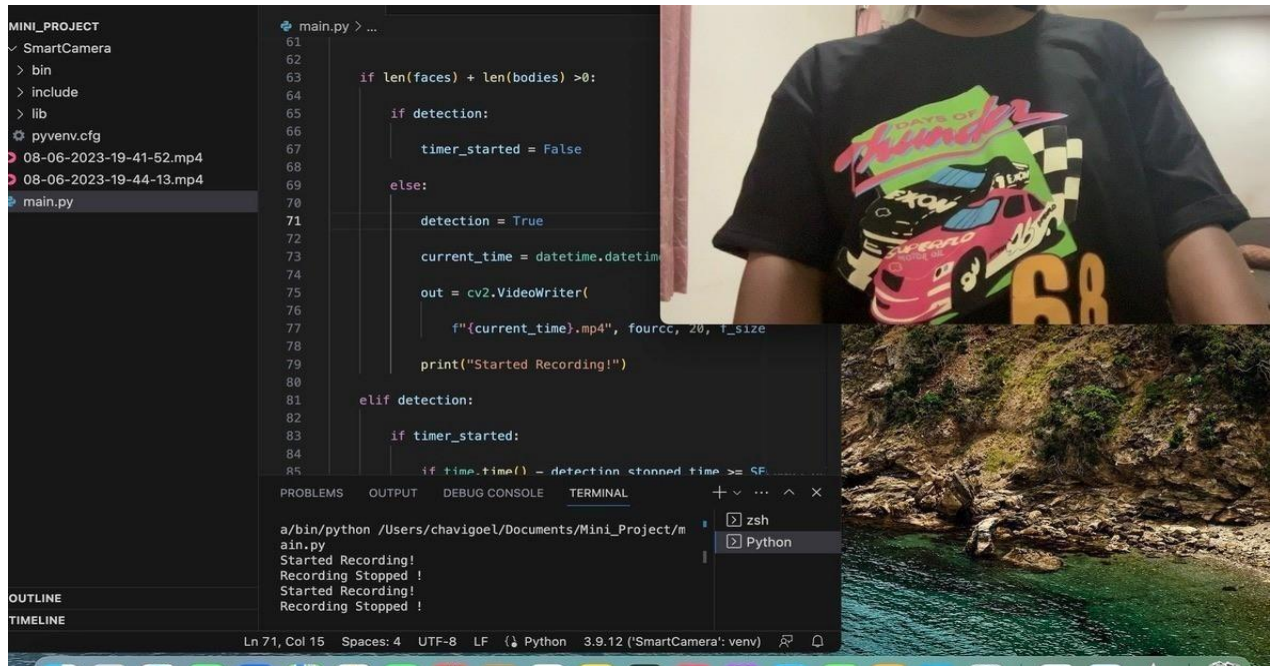
8. SCREENSHOTS



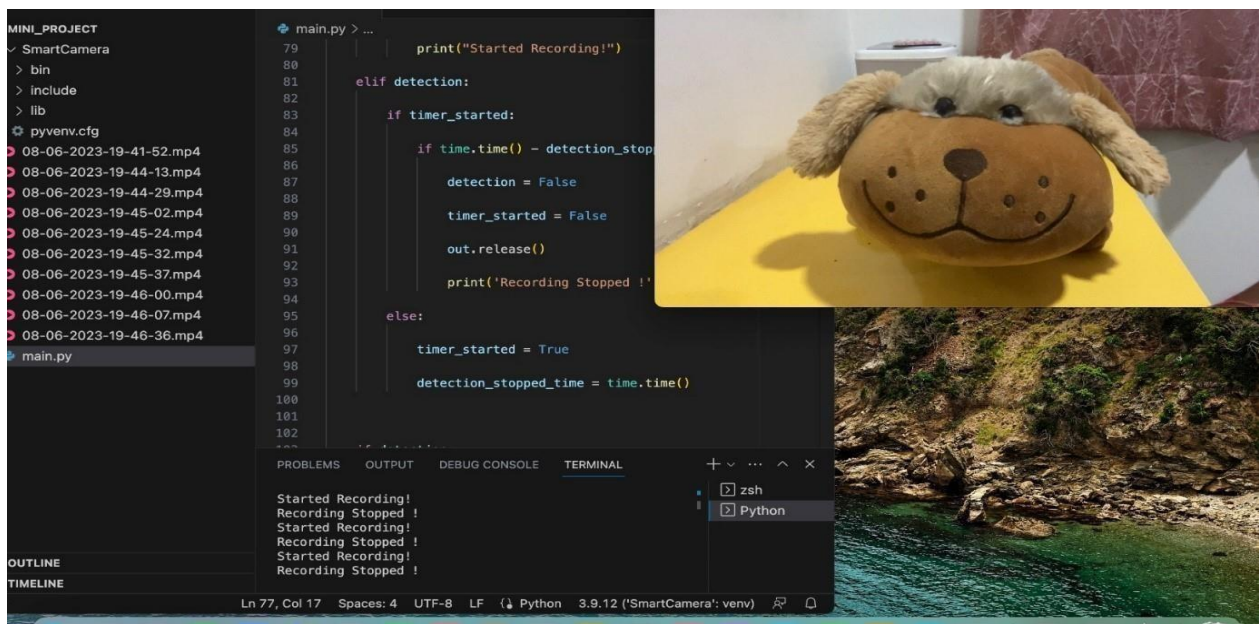
8.1 Person is detected when he is in front of smart camera.



8.2 When there is no person the recording has been stopped and the video



8.3 full body detection the body is getting detected



8.3 In case of animals, sometimes it is being detected and mostly its notdetected. Sometimes itis detected due to eyes.

9 IMPLEMENTATION

ALGORITHM

1. Set up the camera.
2. Include Haar Cascade facial and body XML files.
3. Take the frames of the input video and covert into gray scale.
4. If any body or face is detected in the frame
5. Then set detection as true
6. Start recording the video with file name as date and time.
7. Print start recording
8. If detection is false wait for 5 seconds and stop the recording
9. Print Recording stopped !
10. Go to step 4 until 'q' is pressed.

10. CODE

```
import  
t cv2  
import  
t  
time
```

- cv2 module is imported for image processing and performing computer vision tasks and time, date time module provides a function for getting local time and date.

```
cap = cv2.VideoCapture(0)
```

- we use video Capture() function to capture live stream with webcam. We need to create an object of video
- Capture() to capture a video.

```

face_cascade =
cv2.CascadeClassifier      cv2.data.harcascades
"haarcascade_frontalface_default.xml

body_cascade =
cv2.CascadeClassifier      cv2.data.harcascades
"haarcascade_fullbody.xml"

```

- Cascade Classifier is used to load a .xml classifier file. Frontal face and full body are used for face and body detection.

```

face_cascade =
cv2.CascadeClassifier      cv2.data.harcascades
"haarcascade_frontalface_default.xml

body_cascade =
cv2.CascadeClassifier      cv2.data.harcascades
"haarcascade_fullbody.xml"

```

- Detection Variable is declared to check whether the face or body is detected or not
- Detection stopped time is used to store the recording stopped time and timer started is used whether recording started or not.

```
f_size = (int(cap.get(3)),
int(cap.get(4))) fourcc =
cv2.VideoWriter_fourcc(*"m
p4v")
```

- We write rest of the code in while loop, for repetitive checking of frames and recording the video. We use read() function to read a frame from the video and store that in 'frame' variable. This frame is converted into gray scale. Grayscale simplifies the algorithm and reduces computational requirements.
- read() function will return 2 objects, but we are interested only in the latest one, that is the last frame from the camera. So we are ignoring the first object _, and naming the second as frame, that we will use to feed our detect() function in a second.

```
faces = face_cascade.detectMultiScale(gray, 1.3, 5)
bodies = body_cascade.detectMultiScale(gray, 1.3, 5)
```

- Multi Scale detects objects of different sizes in the input image and returns rectangles positioned on the faces. The first argument is the image, the second is the scale factor (how much the image size will be reduced at each image scale), and the third is the min Neighbors (how many neighbors each rectangle should have).
- The values of 1.3 and 5 are based on experimenting and choosing those that worked best.

```

    if len(faces) + len(bodies) > 0:
        if detection:
timer_started = False
else:
        detection = True                current_time =
datetime.datetime.now().strftime("%d-%m-%Y-%H-%M-%S")

        out =
cv2.VideoWriter(                        f"{current_time}.mp4
", fourcc, 20, f.size)
        print("Started Recording!")

```

- In each frame we check either any body or face is detected. If frame contains face or body is detected, we set the detection variable as true and start saving the recording with the name with current date and time.
- The current date and time is obtained using now() and in the mentioned format as date-month year-hours-minutes-seconds.
- Video Writer() object will create a frame of the video input and store with the file name current time. It take four inputs file name, force, fps and frame size.
- A print statement is there, which prints recording has been started on the console.
- If the detection is already true, we make the time started variable as false

```

elif
detection:if
timer_started:
        if time.time() - detection_stopped_time
            >=SECONDS_AFTER_DETECTION:detection = False
            timer_started =
            Falseout.release()
print('Recording Stopped !')

```

```
timer_started = True
detection_stopped_time = time.time()
```

- If there is no body or face detected and detection is true, we enter into a false loop. Where we check for the timer has been started or not. If timer started is true, we check with present time. If the time is more than 5 seconds we make detection and timer started as false and stop the recording.
 - We print a statement that recording has been stopped.
- If there is no body or face detected, detection is true and timer started is false, then we make timer started as true and note down the detection stopped time in a variable.

```
if detection:
    out.write(frame) cv2.imshow("Smart
Camera", frame)
```

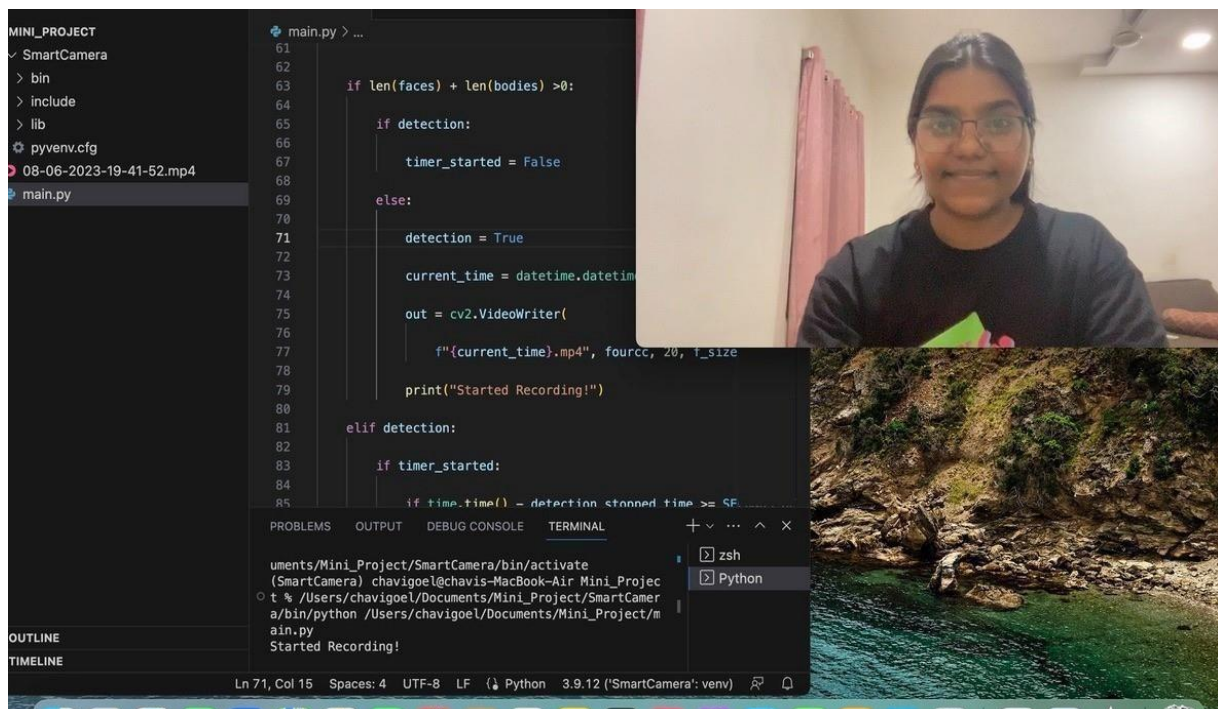
- In this part of code frames are added to recording when the detection is true. cv2.imshow() method is used to display an image in a window and window is named as smart camera.

```
if cv2.waitKey(1) == ord('q'):  
    break  
out.release()  
cap.release()  
cv2.destroyAllWindows()  
)
```

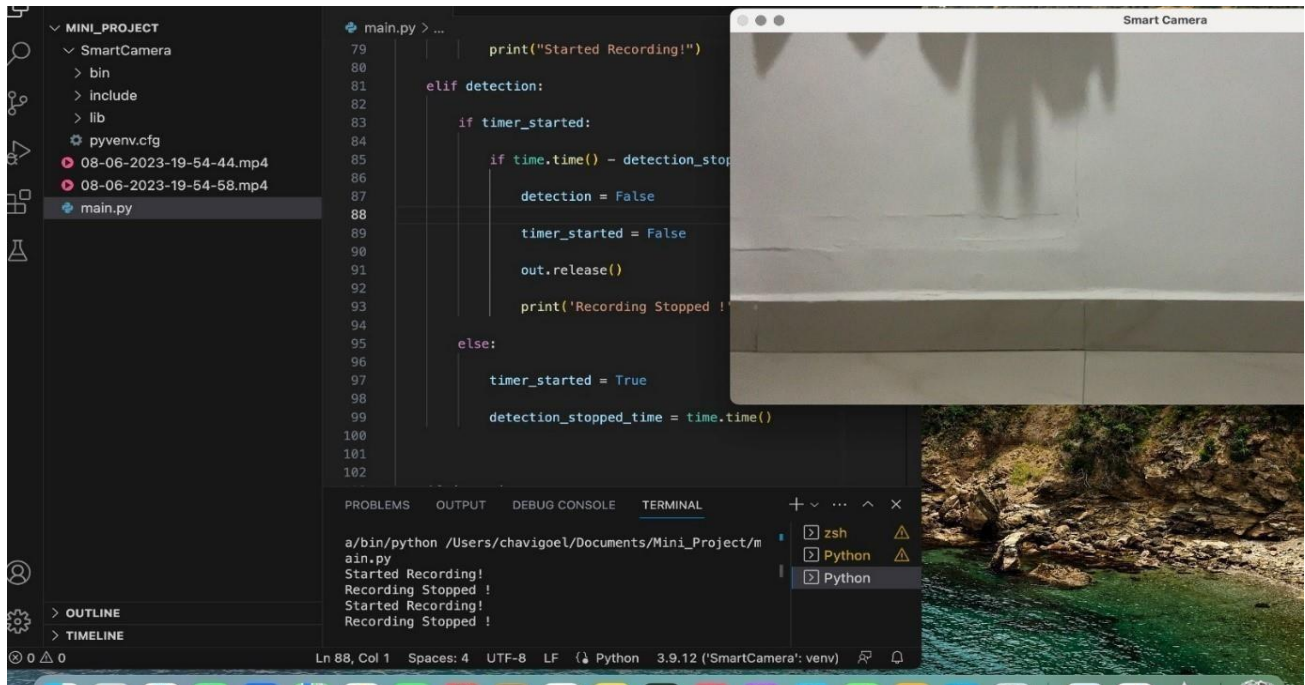
- This if statement checks the key entered by the user. If the user enters character 'q', the execution ends as there is a break statement and comes out of while condition. We use wait key() function to pause frame in the video. This piece of code will stop the program when you press q on the keyboard. We just need to turn off the camera and close the window when we stop the program .So we use release() function to turn off the camera and destroy All Windows() to close the window.

11. TESTING

The purpose of testing is to discover errors. Testing is the process of trying to discover every conceivable fault or weakness in a work product. It provides a way to check the functionality of components, sub-assemblies, assemblies and/or a finished product. It is the process of exercising software with the intent of ensuring that the Software system meets its requirements and user expectations and does not fail in an unacceptable manner. There are various types of test. Each test type addresses a specific testing requirement.

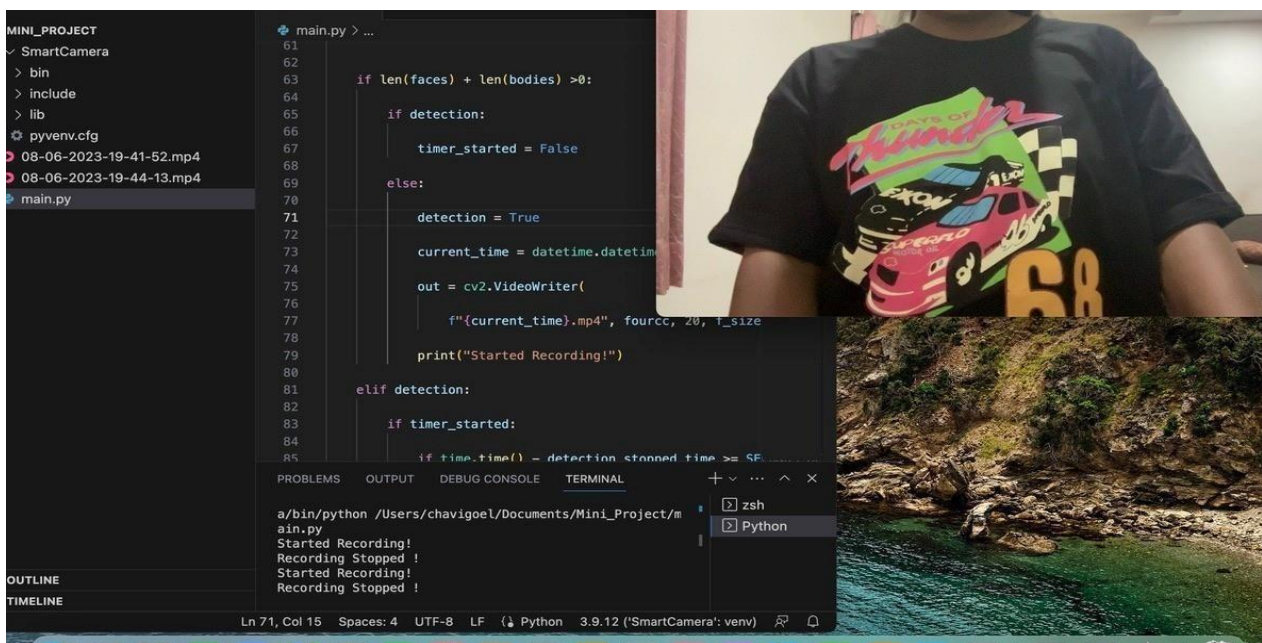


- Person is detected when he is in front of smart camera. You can see the recording has been stored with date and time. In the terminal we got a notification that recording has been started.

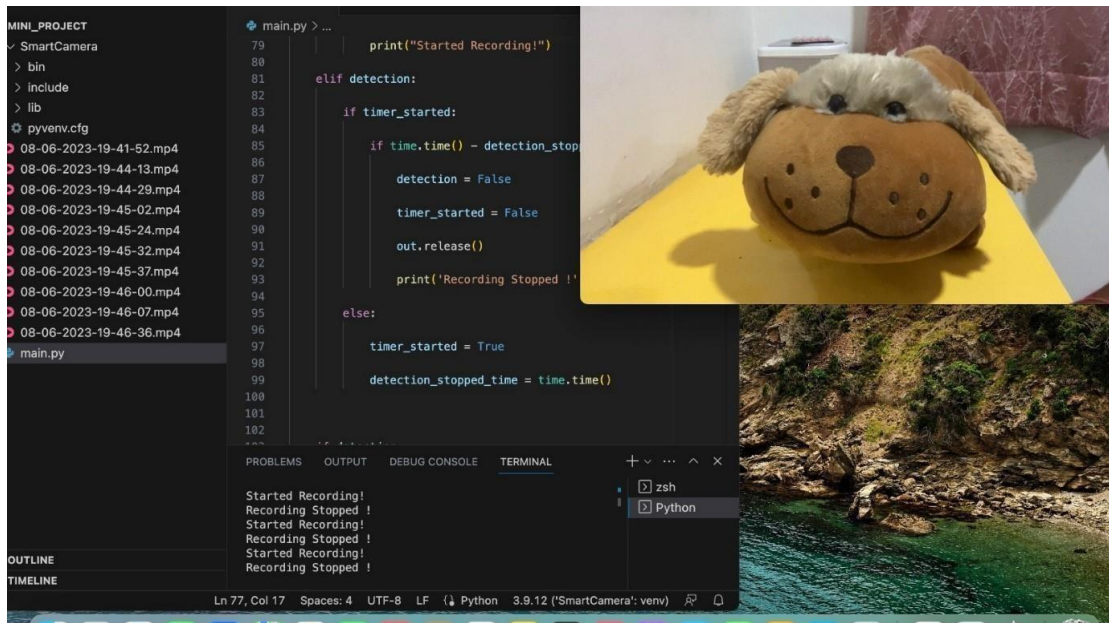


- When there is no person the recording has been stopped and the video is saved in the local folder.

We can see a notification in terminal saying the recording has been stopped.



- As u can see the person face is not there in the camera vision, yet the recording has been started, as we used full body detection the body is getting detected.



- In case of animals, sometimes it is being detected and mostly its not detected. Sometimes itis detected due to eyes.
- Also identifies a person with mask.
- If we show a picture of a person also it identifies and starts recording.
- Body detection is not so accurate.

Mostly cant detect animals , but in few cases due to eyes it is detected as a human

Camera Components	Physically inspect the camera for any defects or damage. Verify that the lens, sensor, and other components are functioning properly
Connectivity	Test the camera's connectivity options, such as Wi-Fi, Ethernet, or USB, to ensure reliable communication with other devices or networks
Power Supply	Test the camera's power source to ensure it powers on/off correctly and remains operational without issues

11.1 Table : Hardware Testing

Software Testing	
Firmware Update	Check if the camera's firmware is up to date. If not, update it to the latest version to ensure optimal performance and security.
Configuration Settings	Test the camera's configuration settings, including resolution, frame rate, motion detection sensitivity, and other parameters, to ensure they can be adjusted according to requirements.
User Interface	Evaluate the user interface of the camera's software, including the setup wizard, menu navigation, and accessibility features, to ensure ease of use.
Remote Access	Test remote access features, such as mobile apps or web interfaces, to verify that users can monitor and control the camera from anywhere with an internet connection.

11.2 Table : Software Testing

Functionality Testing:	
Person Detection	Test the camera's ability to detect persons accurately in various lighting conditions and environments. Verify that it can distinguish between persons and other objects.
Video Recording	Test the camera's video recording functionality, including starting, stopping, and scheduling recordings. Verify that recordings are saved properly and can be accessed later for playback.
Storage Management	Test storage management features, such as local storage options (e.g., SD card, NAS) and cloud storage integration, to ensure efficient utilization of storage space and reliable data retention.

11.3 Table: *Functionality Testing*

Integration Testing:	
Third-party Integration	Test integration with other systems or devices, such as video management software, home automation platforms, or security systems, to ensure seamless interoperability.
API Testing	If applicable, test the camera's API (Application Programming Interface) to verify that it can be integrated with custom applications or services.

11.4 Table: *Integration Testing*

Performance Testing:	
Processing Speed	Test the camera's processing speed by simulating multiple person detection events and recording scenarios to ensure it can handle the workload effectively
Video Quality	Evaluate the quality of recorded video footage, including resolution, clarity, and frame rate, to ensure it meets the desired standards.

11.5 Table : *Performance Testing*

Security Testing:	
Data Encryption	Test data encryption mechanisms to ensure that video footage and othersensitive information are protected during transmission and storage.
Access Controls	Test access controls, such as user authentication and authorizationmechanisms, to prevent unauthorized access to the camera and its data

11.6 Table : *Security Testing*

Usability Testing:	
User Feedback	Gather feedback from users or beta testers to identify any usability issues or areas for improvement in the camera's design or functionality
User Documentation	Review user manuals, help guides, and other documentation to ensure they provide clear instructions for setting up and using the camera effectively.

11.7 Table : *Usability Testing*

USER TRAINING

- Provide thorough training sessions for users on how to operate the smart camera system effectively.
- Cover topics such as basic functionality, troubleshooting common issues, interpreting alerts, and accessing support resources.
- Consider offering both in-person and online training options to accommodate different learning preferences and schedules.

MAINTENANCE

Establish regular maintenance schedules to ensure the smart cameras are functioning optimally. Include tasks such as cleaning camera lenses, checking for firmware updates, and inspecting hardware for any signs of wear or damage. Document maintenance procedures in a comprehensive guide for easy reference by maintenance staff.

TESTING STRATEGY

Develop a testing plan that covers various aspects of smart camera functionality, including image quality, motion detection, object recognition, and network connectivity. Incorporate both manual testing procedures and automated tests where feasible to ensure thorough coverage. Define test cases that simulate real-world scenarios to validate the camera's performance in different environments and lighting conditions. Conduct regular testing of new features or software updates to identify any potential issues before deployment. Establish a feedback loop for users to report any issues or suggestions for improvement, and incorporate this feedback into future testing cycles.

12. CONCLUSION

- With this project, we can increase the surveillance system at a very low cost and easy implementation. It also reduces the storage by capturing only the necessary things. Smart camera can be used almost everywhere.
- In conclusion, the implementation of a smart camera that leverages Artificial Intelligence (AI) and Machine Learning (ML) solely for detecting human movements has shown great potential in enhancing security, efficiency, and convenience
- By focusing on human movements, the smart camera system that accurately distinguish between relevant events and background noise, reducing false alarms and ensuring real- time responses o potential threats and incidents.
- In conclusion, smart cameras represent a groundbreaking fusion of cutting-edgetechnologies and advanced capabilities, reshaping the way we perceive, capture, and analyse visual information. These intelligent devices are equipped with artificial intelligence, machine learning, computer vision, and internet connectivity, enabling them to excel in various applications and industries.
- The future scope of smart cameras is highly promising, with numerous potential advancements and applications on the horizon. From revolutionising transportation through autonomous vehicles to transforming industries like healthcare, retail, and manufacturing, smart cameras will continue to play a pivotal role in shaping a more connected, efficient, and secure world.

13. REFERENCES

- A. **Brands and Manufacturers:** Look for well-known brands and manufacturers in the smart camera industry, such as Nest, Arlo, Ring, Wyze, or Eufy. These companies often offer advanced motion detection features.
- B. **Product Descriptions:** Check the product descriptions and specifications of smart cameras to see if they specifically mention human detection or person recognition capabilities.
- C. **Customer Reviews:** Reading customer reviews can provide valuable insights into how well a smart camera performs in recognising and recording human movements.
- D. **Tech Blogs and Reviews:** Check out technology blogs and review websites that specialise in smart home products. They often evaluate and compare different smart cameras for their features, including human detection capabilities.
- E. **Online Retailers:** Popular online retailers like Amazon, Best Buy, and others usually have customer reviews and product descriptions that can help you identify smart cameras with human detection features.

14. GANN CHART
HUMAN DETECTION USING SMART CAMERAS

PROCESS	WEEKS													
	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Abstract														
Literature/ Requirement gathering														
Design/ prototype														
Execution/ Project development														
Testing the project														
Report Generation														